Philip Robinson

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https://github.com/probinso

10+ years experience in software, machine learning & DevOps on interdisciplinary teams supporting work across natural sciences, computer securities, remote sensing &|| citizen sciencery

Skills/Experience

Machine Learning	Remote/Global Teams	Linux Systems	MLOps/DevOps
Data Sciences	Technology Transfer	Education/Mentoring	Rubber Ducking

Technologies

Python/PyTorch/sklearn	Angular/Next.js/FastAPI	scipy.signal/librosa/sox	SQL/BQL/PostGIS
C/C++	GDAL/cartopy/ESA-SNAP	AWS/GCP/firebase	Linux/Bash/Docker

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Blue Oasis — **ML Engineer (Digital Twins)** — **Contract · May 2025 – Oct 2025**

- Improved ML edge multi-label acoustic event detection pipeline, for tracking vessels & marine life
- Full stack + security & performance on HydroTwin, a user focused acoustics analytics/IR dashboard
- Advised DL & surrogate model design on PHAROS & SmartFisher, for ecosystem monitoring & simulation

Global Fishing Watch — **Scientific Programmer (Earth Imaging)** — **May 2022 – March 2025**

- Implemented remote sensing pipeline for detecting vessels & matching to reported GPS locations
- Evaluated vessel detection over terrabytes of processed synthetic aperture radar (SAR) images
- Developed gfwsat to join & catalog millions of satellite scenes, serving global ocean surveys
- Developed simple offline satellite image labeler, supporting all shapely geometries by lat/lon
- Led technology transfer work, moving research prototypes to production & automated pipelines
- Developed collaboration strategies & best practices docs, for remote & international teams

GrammaTech — **Deep Learning Researcher II (Security)** — **March 2020 – April 2022**

- Research on binaries in vulnerability detection & improving code legibility from decompilation
- Developed feature extraction pipeline over local cluster & virtual machines on terrabytes of data
- Implemented, verified & incorporated ML/DL code from academic research into release products
- Participated in authoring & reviewing SBIR/STTR proposals for DOD/DARPA funding solicitations

HappyWhale — **Mobile & Full Stack Engineer** — **Contract · March 2019 – March 2020**

- Developed x-platform citizen science phone app for eco-tourism & wildlife population surveys
- Developed core tools for tracking marine life sightings & individual whale identification
- Contributed to data access API & web UI, supporting researchers in population ecology studies

NASA's Jet Propulsion Lab — **Data Science Intern (Natural Text)** — **June 2018 – Sept 2018**

- Prototyped employee expert recommender system, to eliminate weeks in ticket triage/assignment
- Designed & completed prototype from research papers & advisement of top NASA/JPL employees

OHSU Neuroimage Lab — **Graduate Research Assistant (Medical)** — **Oct 2017 – June 2018**

- Analyzed relationship of adjacent microbiome populations for menopause & reproductive health
- Developed processing/audit tools for survey & fMRI data to support ABCD study for ADHD/ASD
- Provided git, project management & security trainings for teams with varying readiness levels

OHSU MSc Research — **Machine Learning (Marine Acoustics)** — **Sept 2017 – Dec 2019**

- Developed track explorer for 10YR continuous audio track, supporting Aloha Cabled Observatory
- Developed anomaly detectors for noisy marine acoustics, based on variational auto-encoders
- Measured effects of spectral subtraction & DL models to reduce noise & improve audio quality

Galois Inc. — **R&D Software Engineer (Security)** — **April 2014 – Dec 2015**

- Developed processing pipelines on local clusters for evaluator workflows with multiple SBIR/STTRs
- Setup brittle PPAML languages on 30+ participants' devices for two annual DARPA summer schools
- Co-authored secure internet voting feasibility study for non-technical, policy-focused audience

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Oregon & Health Science University
Computer Science & Machine Learning
Masters of Science

Western Washington University
Computer Science & Cryptography
Bachelor of Science, Math Minor